Current asset 1 / v0.4 figure1A | equal native width

CURRENT

Local opportunity does not ensure end-to-end gain

Observed case	Favorable local measurement		Realized outcome
M4 · HumanEval30 Preregistered, fixed k = 2	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→	● **Exact + profitable** 1.387× · 90/90 exact
M4 · representative 24Q Frozen representative study	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→	● **Exact + negative** 0.842× · 72/72 exact
RTX 5070 Ti Laptop CUDA batched verification	$V_2^{\text{micro}} = 0.983–1.009$ Near one target-decode step	→	□ **Correctness blocked** No admissible speedup
358H · NPU draft Qualified Phase-3 path	48.08 vs 30.81 tokens/s NPU vs CPU · isolated, 8,192 tokens	→	● **Exact + negative** 0.352× · 90/90 exact

Categorical cases, not a hardware ranking. Filled = qualified; open = blocked. Details and CIs: Figures 2–4.

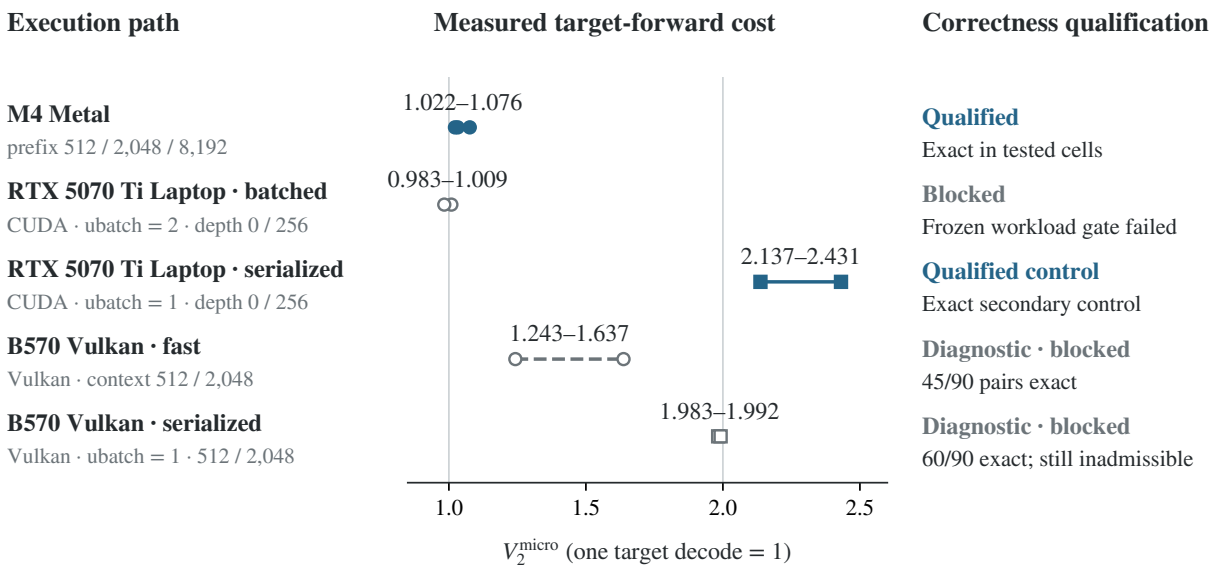
Micro cost ≠ complete target work; isolated throughput ≠ stateful cost. Speedups cover post-prefill decode.

v0.4

Case	Local indicator	Realized outcome
M4 HumanEval30, k = 2	V_2^{micro}: 1.022–1.076	Profitable: 1.387×; 90/90 exact
M4 representative 24Q	V_2^{micro}: 1.022–1.076	Negative: 0.842×; 72/72 exact
RTX 5070 Ti Laptop, batched	V_2^{micro}: 0.983–1.009	Blocked; no admissible speedup
358H NPU draft	NPU/CPU: 48.08 vs 30.81 tokens/s	Negative: 0.352×; 90/90 exact

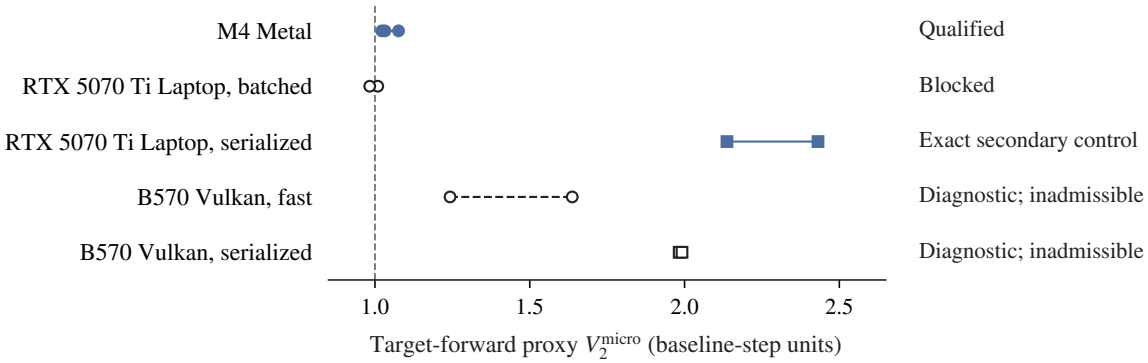
Current asset 2 / v0.4 verification | equal native width

CURRENT



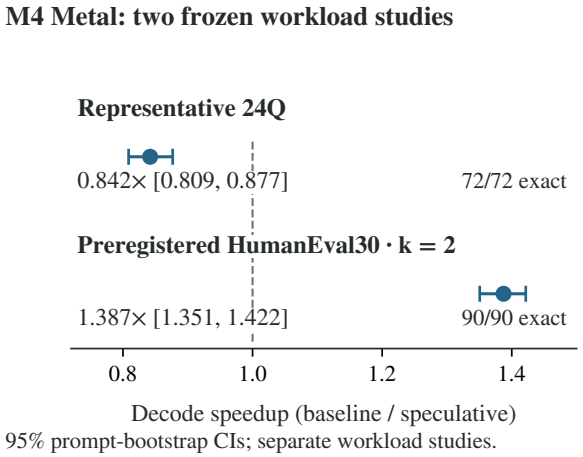
Filled = qualified; open = blocked/diagnostic. Circles = batched/fast; squares = serialized.
Ranges span tested contexts (not CIs); contexts are not fully matched. $V_2^{micro} \neq R_{target}$.

v0.4

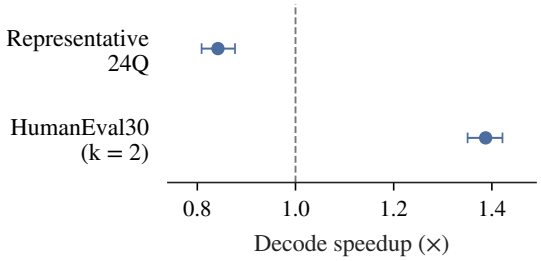


Current asset 3 / v0.4 m4 | equal native width

CURRENT

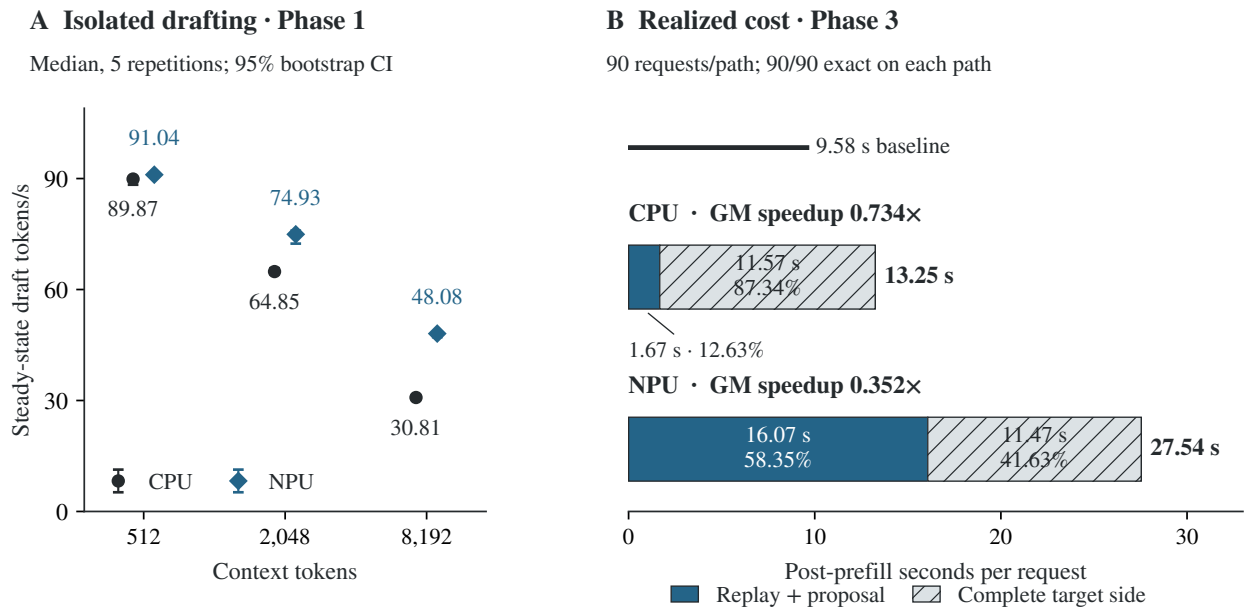


v0.4



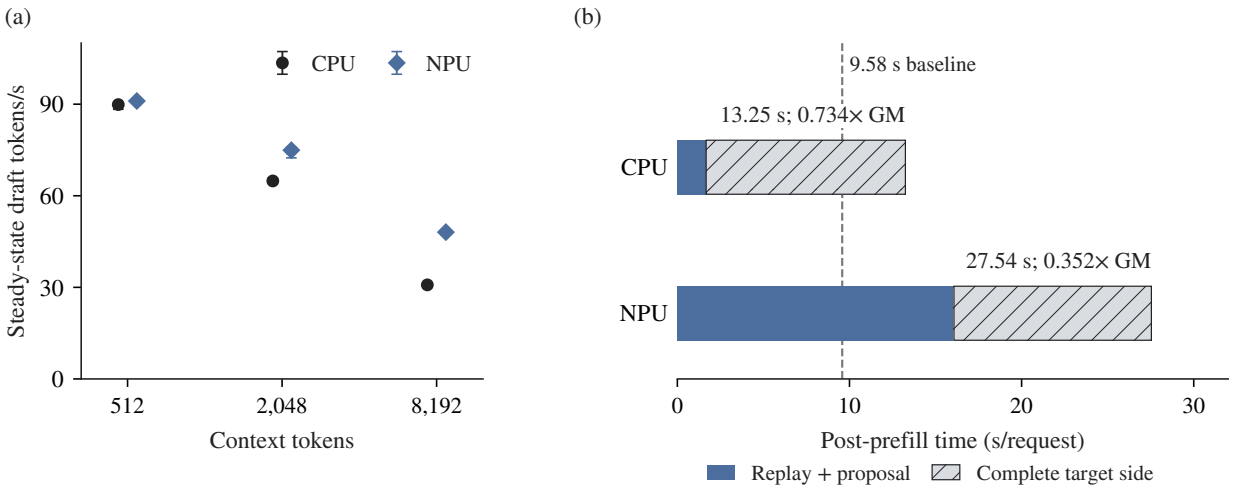
Current asset 4 / v0.4 state | equal native width

CURRENT



Residual: CPU 4.81 ms (0.04%); NPU 5.06 ms (0.02%). Bars: arithmetic means; speedups: prompt GMs.
Replay and proposal share one timed call. Phase-1 steady-state throughput excludes the full-prefix state cost.
Speedup 95% CIs: CPU [0.693, 0.771]; NPU [0.333, 0.369]. Initial prompt prefill is excluded from Phase 3.

v0.4



CURRENT

Categorical regimes under the frozen correctness contract

Mechanism	Not admissible	Correctness-qualified	
	No admissible speed claim	Negative ($S < 1$)	Profitable ($S > 1$)
Workload conditioning		● **M4 representative 24Q [P]** 72/72 exact	● **M4 HumanEval30 [P]** 90/90 exact
Verification constraint	○ **RTX batched [M]** Cheap micro path; gate fails	● **RTX serialized [S]** CPU→RTX / RTX→RTX; exact	
Correctness barrier	○ B570 fast: 45/90 [D] ○ B570 serialized: 60/90 [D]		
Target-side cost		● **358H CPU [P/M]** 90/90; target side 87.34%	
State-interface cost		● **358H NPU [P/M]** 90/90; replay + proposal 58.35%	

P primary · S secondary · M mechanism · D diagnostic. Fractions count exact token-ID pairs.

v0.4

Case	Qualification	Outcome	Class	Case annotation
M4 representative 24Q	Qualified; 72/72 exact	Negative	P	Workload-conditioned
M4 HumanEval30	Qualified; 90/90 exact	Profitable	P	Workload-conditioned
RTX batched	Blocked; gate failed	Not admissible	M	Verification-constrained
RTX serialized controls	Qualified; 90/90 each exact	Negative	S	Verification-constrained
B570 fast	Inadmissible; 45/90 exact	Not admissible	D	Correctness-blocked
B570 serialized	Inadmissible; 60/90 exact	Not admissible	D	Correctness-blocked
358H CPU	Qualified; 90/90 exact	Negative	P/M	Target-side dominated 87.34% target side
358H NPU	Qualified; 90/90 exact	Negative	P/M	State-interface bound 58.35% replay + proposal